

Lab Task 1: Introduction to Python & NLP

Aim:

To get familiar with Python programming and practice basic Natural Language Processing (NLP) operations using libraries such as NLTK and spaCy.

Code:

```
import nltk

import spacy

nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('stopwords')
nltk.download('averaged_perceptron_tagger')
nltk.download('averaged_perceptron_tagger_eng') # fix

nlp = spacy.load("en_core_web_sm")

text = "Natural Language Processing is a part of Artificial Intelligence."

from nltk.tokenize import word_tokenize

tokens = word_tokenize(text)

print("Tokens:", tokens)

from nltk.corpus import stopwords

filtered = [w for w in tokens if w.lower() not in stopwords.words('english')]

print("Filtered:", filtered)

# POS tagging (now works)

print("POS:", nltk.pos_tag(tokens))

doc = nlp(text)

for token in doc:

    print(token.text, "->", token.pos_)
```

OUTPUT:

Tokens: ['Natural', 'Language', 'Processing', 'is', 'a', 'part', 'of', 'Artificial', 'Intelligence', '.']

Filtered: ['Natural', 'Language', 'Processing', 'part', 'Artificial', 'Intelligence', '.']

POS: [('Natural', 'JJ'), ('Language', 'NNP'), ('Processing', 'NNP'), ('is', 'VBZ'), ('a', 'DT'), ('part', 'NN'), ('of', 'IN'), ('Artificial', 'NNP'), ('Intelligence', 'NNP'), ('.', '.')]

spaCy Tokens and POS Tags:

Natural -> PROPN

Language -> PROPN

Processing -> PROPN

is -> AUX

a -> DET

part -> NOUN

of -> ADP

Artificial -> PROPN

Intelligence -> PROPN

. -> PUNCT